

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I Maddirala Gopi Krishna Reddy 192372055(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Type-1 vs Type-2 Hypervisors for mission-critical cloud datacenters

Architectural and Operational Comparison

Factor	Type-1 (Bare-metal/Native)	Type-2 (Hosted)
Architecture	Runs directly on bare metal hardware. Acts as the operating system	Runs as an application on top of an existing host OS
Performance	Direct access to CPU, memory and PCIe device Ultra-low overhead	High overhead due to double scheduling and system call translation
Security	Minimal attack surface	High attack surface. Vulnerabilities in host OS directly compromise all VMs
Manageability	Centralized enterprise management interfaces supporting hot migration and automated failover	Individual application management; lacks enterprise clustering, live migration and dynamic resource
Examples	VMware ESXi, KVM, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation, Parallels

Justification of the Better choice:

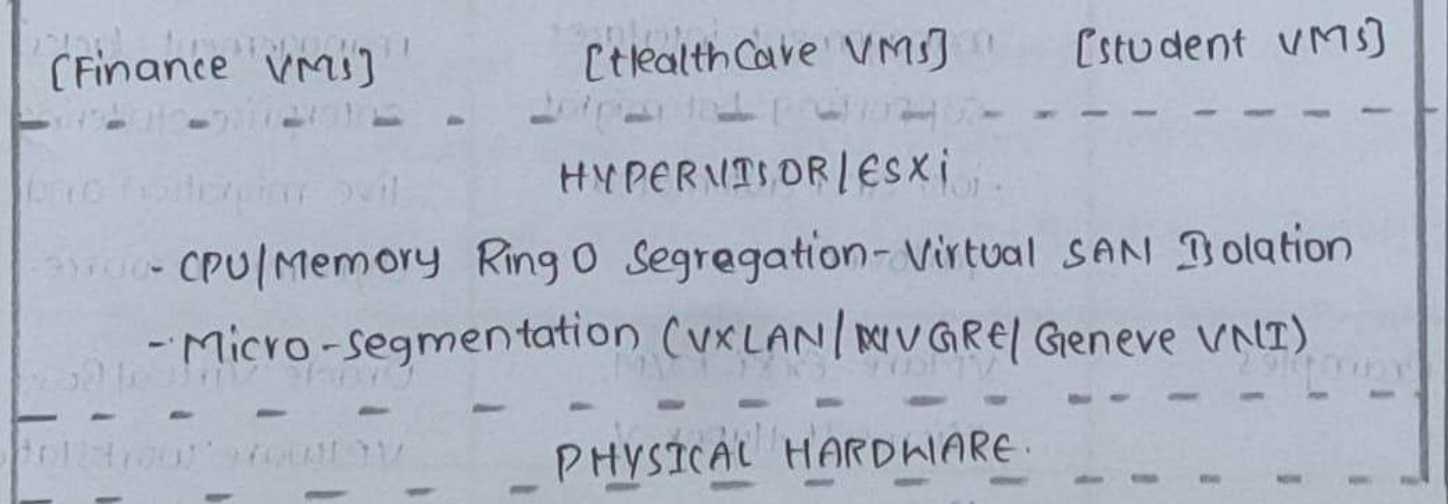
Type-1 is the definitive choice for mission critical cloud workloads.

1. Performance and determinism: High transaction applications require predictable latency. Type-1 avoids the host OS scheduler

-ling bottlenecks present in Type-2
Hard Isolation and Security: High-Value data mandates hardware-level ring 0 isolation. A host OS exploit in Type-2 would grant root access to all tenant VMs.

High Availability: Type-1 hypervisors natively support live migration automated load balancing (DRS) and instant VM fault tolerance without guest OS downtime.

2. Multi-Tenant Workload Isolation and Vulnerability Analysis: Virtualization Isolation Architecture: To safely host multi-tenant workloads with strict regulatory compliance on shared physical hardware, the virtualization stack implements isolation across three main layers:



Compute Isolation: Memory virtualization uses nested page tables (NPT/Intel EPT) to map guest physical addresses directly to host physical addresses, ensuring tenant VMs cannot access adjacent VM RAM space.

Network Isolation:- Overlay networks Using VXLAN/Geneve VNIs (Virtual Network Identifiers) enforce Strict L2/L3 Separation. Micro-Segmentation Via distributed Firewalls prevents east-west lateral movement.

Storage Isolation:- Logical Unit Number masking, Storage Virtualization tags and encrypted datastores ensure a tenant's IOPS quota and Storage domain remain strictly private.

Two Major Attack Surfaces:-

1. Hypervisor Escape (VM Escape):- An attacker exploits a bug in the Hypervisor Virtual hardware emulation to execute Code directly on the host hypervisor Context, granting total Control over all Co-located VMs across tenants.
 2. Micro Architectural Side-channel Attackers:- Rogue VMs utilize hardware-level CPU CPU branch prediction, Speculative execution and shared CPU Cache to extract Sensitive data from neighbouring tenant VMs running on the Same Physical CPU Core.
3. Host Utilization Bottleneck Analysis and Corrective Actions:-
Metric Analysis and Bottleneck Identification:-

Metric	Measured value	Status	Criticality
CPU Utilization	92%	High (>80%)	CPU Contention/ Ready time Spikes

Memory Utilization	88%	High (>80%)	Nearing hypervisor Swapping boundary
Storage I/O Latency	35ms	Severe Bottleneck (Target: <10ms)	High disk queue depth / IOPS Saturation
Average VM boot Time	170s	Serve Delay	Symptom of I/O and CPU Starvation

Primary Bottleneck: Storage I/O Subsystem. Storage latency at 35ms is the primary system bottleneck. High latency causes VM Operating Systems to stall waiting for disk read/write acknowledgments, which cascades into prolonged VM boot times (170s) as system binaries take minutes to load.

Secondary Bottleneck: Compute and memory Overcommit. A 92% CPU load combined with 88% RAM allocation indicates that the host physical capacity is saturated. The CPU is likely spending significant time handling I/O wait cycles and hypervisor context switching rather than executing actual workload instructions.

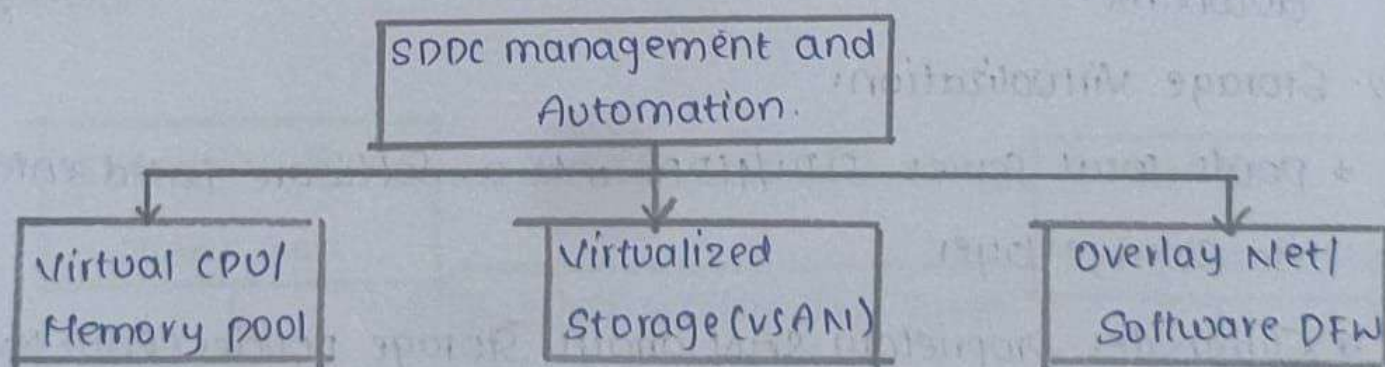
Immediate Correction Actions:

1. **Storage Subsystem:** Migrate high-I/O VMs to an NVMe/SSD tier or distributed SAN with higher IOPS capability. Using Storage Live Migration. Check for alignment issues, active snapshot buildup or rogue storage-heavy background jobs.

2. Compute and RAM Balancing: Invoke Dynamic Resource Scheduler (DRS) or manually live-migrate high-consumption VMs to underutilized host nodes in the cluster. Verify CPU Ready Time if high, reduce the number of virtual CPUs (vCPUs) allocated per VM to decrease scheduling wait queues.
3. Boot Optimization: Enable Host Caching/Read Caching to accelerate initial OS boot sequences without straining the primary storage array.

4. SDDC vs Traditional Datacenter Agility Comparison

Software-Defined datacenter (SDDC) decouples management and control planes from underlying physical hardware, implementing abstraction across Compute, Storage and Networking.



Strategic Comparison

Dimension	Traditional DataCenter	Software Defined DataCenter
Provisioning Speed	Days to weeks (manual hardware racking, cabling, VLAN tagging)	Seconds to minutes via API calls, Infrastructure as Code and Orchestration workflows

Resource Allocation	Static, Siloed hardware pools. poor utilization	Dynamic policy-driven allocation, auto-scaling and global pool extraction
change Management	Manual Switch Configurations SAN LUN masking, ticket queues	Programmatic policy enforcement through declarative templates

core SDNC Considerations:

1. Computer Virtualization:

- * Converts physical Servers into unified Compute pools.
- * Enables instant provisioning, automated workload rebalancing and transparent host maintenance without application downtime.

2. Storage Virtualization:

- * Pools local server SSDs/HDDs into a Software based enterprise storage layer.
- * Eliminates proprietary SAN arrays. Storage policies (RAID levels) are attached directly to VMs via Software policies rather than hardware LUN reconfiguration.

3. Network Virtualization:

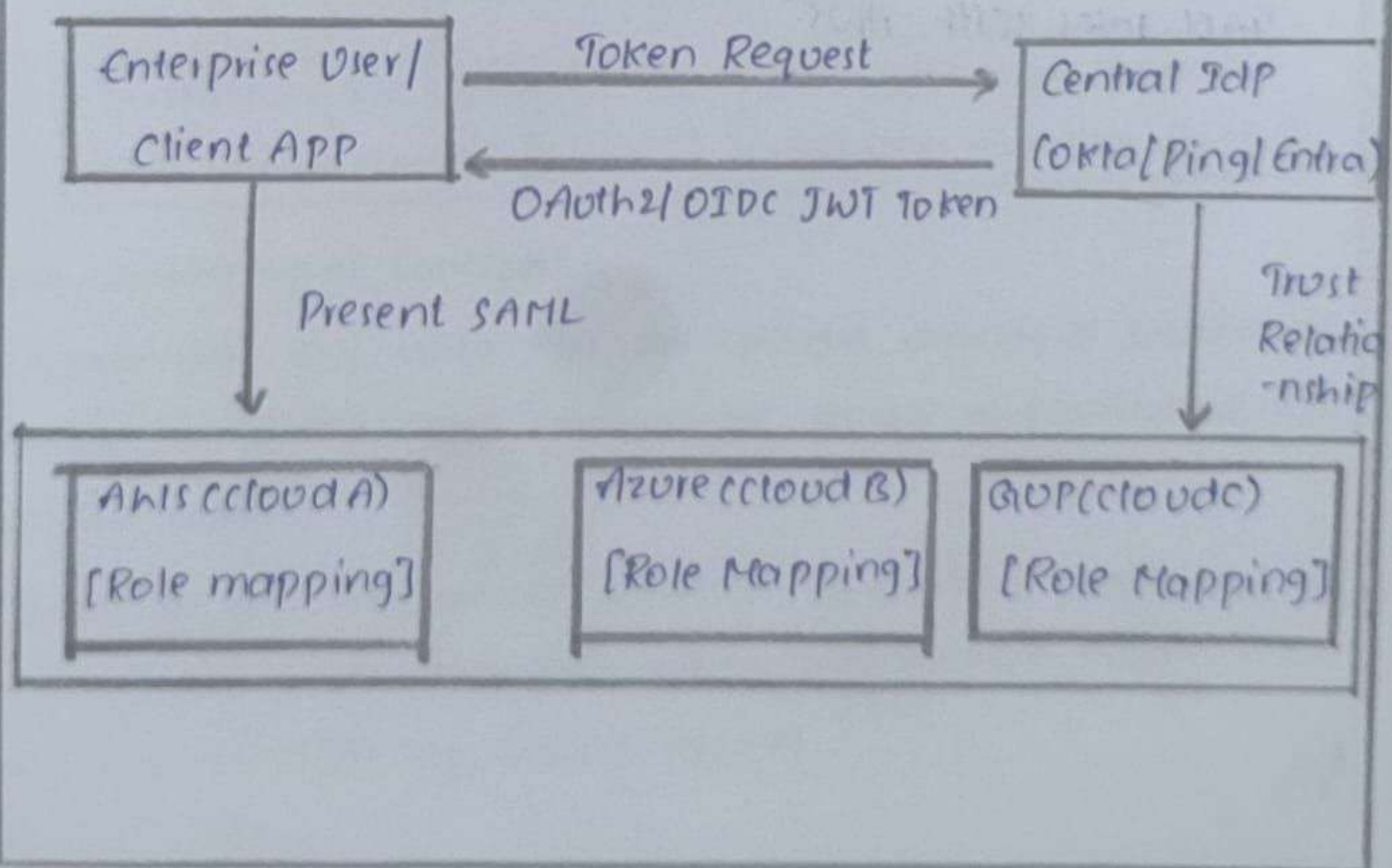
- * Decouples logical Servers from physical Switches using VXLAN/ Geneve overlay networks.
- * Allows Creation of vswitches, Virtual routers, load balancers and firewalls entirely in Software.

5 Multi-cloud Federation Identity and privacy Architecture

Issue Analysis: The failure in the multicloud federation project stems from Identity Silos and Schema Mismatches. Different cloud providers use incompatible user identity attributes, assertion formats and trust boundaries. Without standard protocols mapping users across cloud domains leads to access denials, broken authorization contexts, and privacy leaks.

Secure Federation and privacy Architecture

To solve this, implement a standardized federated Identity Management (FIM) System utilizing SAML 2.0, Open ID Connect (OIDC) and OAuth 2.0 anchored around a centralized identity provider.



Architecture Components

1. Centralized Identity Provider: Establish a Single Source of truth that holds master identities and handles authentication requests across enterprise identity stores.

2. Federated Standards:

- * Use OIDC for modern API applications.
- * Use SAML 2.0 for enterprise web portal integrations.
- * Rely on SCIM to automatically provision users across cloud accounts in real time.

3. Attribute mapping layer:

Implement Attribute-Based Access Control (ABAC) to normalize incoming token claims across clouds to provide specific

IAM roles with AWS.